

Explainable AI for Histopathology Cancer Grading

(2022)

Problem

Grading prostate cancer aggressiveness (Gleason grading) from histopathology whole-slide images.

Method

A CNN model with Grad-CAM and attention-based explainability overlays.

Dataset

PANDA challenge histopathology dataset.

Evaluation

Quadratic kappa and a pathologist agreement study.

Results

Model-pathologist agreement reached a quadratic kappa of 0.74, comparable to inter-pathologist agreement in the literature.

Limitations

Explanations are not validated against ground-truth cellular annotations, raising questions about explanation faithfulness.